



Universität Hamburg
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Master Thesis

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

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by

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Abstract

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Nomenclature

Abbreviations

Abbreviation	Definition
USA	United States of America
EU	European Union

Variable Names

Symbol	Definition	Unit
α	greek alpha	[N]
β	greek beta	[J]

1

Introduction

The Hadley cell (HC) is the dominant meridional large-scale circulation in the tropics. Its ascending branch at the equator is collocated with inter-tropical convergence zone (ITCZ), which is characterized by large amounts of precipitation. On the other hand, the dry, semi-arid and arid regions of subsidence in the subtropics are associated with the descending branch of the HC. In the annual mean and in the equinoxes there are two similar overturning cells in the tropics - one in each hemisphere. It extends from 0 to 30 °N in the Northern hemisphere and equivalently 0 to 30 °S in the Southern hemisphere. Throughout the annual cycle there are strong variations in the HC's extent and strength: As the maximum of insolation moves poleward, so does the ITCZ and the HCs, resulting in a strong cross-equatorial winter (hemisphere without the maximum) and a weaker summer HC (hemisphere with the maximum). This seasonal migration is related to the global monsoon, which has been identified as an angular momentum conserving HC regime (Bordoni & Schneider, 2008).

The HC is responsible for the poleward transport of energy and the equatorward transport of moisture, thereby reducing the meridional temperature gradients imposed by solar insolation. The HC has strong implications for human and animal life as well as agriculture and vegetation on this planet. Its extent crucially modulates global precipitation patterns by setting the regions of ascend and descend. The strength of the HC determines what amount of energy is reaching the off-equatorial region and what amount of moisture is reaching the equator, setting the potential precipitation amounts. This is particularly evident in the Sahel region, which is located approximately 10 °N of the equator on the African continent. This semi-arid region receives most of its precipitation in the boreal summer when the HC and the ITCZ migrate northward. Thus, the rainfall in the Sahel is dynamically linked to the HC position. Changes in the HC also affect the livelihood of the local population and economies that remain heavily dependent on agricultural production.

This has also been shown for paleoclimate. Paleoclimate simulations of the HC display large changes in HC strength and extend during the last glacial maximum and the mid holocene due to changes in orbital forcing (D'Agostino et al., 2017). This is also supported by evidence from tropical paleoclimate proxy records of precipitation changes (McGee et al., 2014). This past evidence of strong HC changes combined with the large societal impact of precipitation patterns give a strong reason to investigate the projected HC changes of human driven greenhouse gas emissions in the current era.

The CO₂ in the atmosphere is very uniformly distributed. The forcing that it imposes is not. The forcing is critically set by the difference between surface and stratospheric temperature,

which varies strongest in the meridional direction. Given that the HC is flattening the meridional energy gradient, the response of the HC strength depends on meridional forcing structure. This has been shown for quadrupling of CO₂ only in certain latitude belts in coupled simulations (Kim et al., 2022). A forcing in mid latitudes weakens the Hadley circulation whereas a forcing at the equator leads to it strengthening. We investigate how the *vertical* structure of CO₂ forcing, rather than its horizontal structure, alters atmospheric circulation. We achieve this by running simulations with a General Circulation Model (GCM) while quadrupling CO₂ only in the troposphere and only in the stratosphere using a pre-industrial control tropopause height to separate the two cases. To our knowledge, these experiments have not yet been described by the literature.

Climate simulations from the Coupled Model Intercomparison Project (CMIP), have established a HC weakening and widening under climate change (Lu, Vecchi, & Reichler, 2007; Xia, Hu, & Liu, 2020). Explaining this, previous studies have emphasized the importance of drivers related to the troposphere (Chemke & Polvani, 2021; Fan & Dommenges, 2024). However, historical simulations show that stratospheric forcing also influences the general circulation. In particular stratospheric ozone depletion has led to a cooling of the Southern hemisphere lower-stratospheric polar cap. This has been identified as the main driver for the widening of the Hadley circulation over the second half of the twentieth century (Polvani et al., 2011). The goal of our study is to disentangle the contributions of tropospheric and stratospheric CO₂ to the established HC weakening with uniform CO₂ distribution. We find that the HC response to tropospheric CO₂ is a strengthening and widening in both hemispheres. This opposes our expectations evoked from the literature, which identifies tropospheric warming parameters as the main drivers for HC weakening under uniform CO₂ concentrations. Our results suggest that the weakening of the HC originates from stratospheric cooling caused by stratospheric CO₂. To investigate and explain the different responses of the HC strength we employ angular momentum and energy budget frameworks. We use the axisymmetric angular momentum framework to attribute HC changes to changes in tropopause height, meridional temperature, gradient, and vertical stability. Since this theory ignores the contribution of eddies to HC strength we also investigate changes in eddy divergence to explain the observed changes in the mean circulation. The energetic framework is a diagnostically unifying approach. It uses changes in top-of-atmosphere (TOA) radiative fluxes and moist static energy to distinguish different contributing factors to HC changes.

We contribute to the advancement of the field by revising the applicability of different HC frameworks and simplified models. Splitting up CO₂ forcing in two vertical layers puts existing theories to the test and gives the opportunity to rethink accepted truths in climate science. Overall, this is a contribution to foundational research that advances our physical understanding of the drivers of general circulation changes.

TBD Overview over the study Is this really necessary?

2

Theory

2.1. The atmosphere under greenhouse gas forcing

tropospheric warming Why is the troposphere warming under greenhouse gas forcing? What is the warming structure? How can we explain it? Enhanced upper tropospheric warming in the tropics Arctic amplification

stratospheric cooling Why is the stratosphere cooling under greenhouse gas forcing?

2.2. Hadley circulation strength under greenhouse gas forcing

The HC can be characterized by its strength and extent. The HC can be described by the mass stream function (MSF)

$$\psi_{\max} = \frac{2\pi a \cos \varphi}{g} \int_{p_s}^0 \bar{v} dp'. \quad (2.1)$$

The strength is given by the maximum of the $\psi_{\max}$ and the extent by the latitude at which $\psi_{\max}$ becomes zero. Direct effect of CO₂ forcing on the HC (Bony et al., 2013) Quantifying the change of the HC extent and strength under climate change (Lu, Vecchi, & Reichler, 2007) (Vallis et al., 2015) What is the role of different feedback mechanisms in shaping the HC response to climate change (Feldl & Bordoni, 2016) Evaluation of different drivers of HC changes (Chemke & Polvani, 2021)

Which drivers have been identified for the weakening of the HC

2.3. The angular momentum budget

Considering the angular momentum budget of the Earth we are trying to answer the question why there is a meridional circulation in the atmosphere in the first place. An answer is provided by the axisymmetric theory of Held and Hou (1980). It also provides a scaling for the HC strength and extent linking it directly to thermodynamical parameters. Since this theory ignores eddies, we are further investigating their effect on the HC in the next subsection. Since there is no unifying theory incorporating angular momentum conservation and angular momentum transport by baroclinic eddies, we introduce the energetic budget framework of the HC to explain why it changes. This is a more diagnostic theory which does not explain

the existence of the HC from first-hand principles but rather takes the necessity transport energy poleward as granted. It can be inferred from observational data of the TOA radiative imbalance showing an excess of incoming radiative fluxes relative to the outgoing radiation at the equator and vice versa at higher latitudes. This transport by the the atmosphere can be split up into a mean component, i.e., the HC, and and eddy component.

2.3.1. Axisymmetric theory

The axisymmetric theory considers a flow in the $\varphi - z$ -plane with no meridional circulation ($v = w = 0$). This prevents the existence of large scale barotropic or baroclinic eddies. The questions why there is a meridional circulation in the first place is answered by the axisymmetric theory (Held & Hou, 1980). This question is raised because one could imagine a world where Earth is just in radiative equilibrium everywhere. This means that every point on Earth emits just as much radiation as it receives from the sun. Thus, there would not be any meridional temperature or energy transport necessary. Given a temperature profile of the Earth

$$\frac{\theta_E}{\theta_0} = 1 - \frac{1}{3}\Delta_h(3\sin^2\varphi - 1) + \Delta_v\left(\frac{z}{H} - \frac{1}{2}\right) \quad (2.2)$$

obtained from radiative equilibrium conditions (see Figure 2.1a), one can obtain a zonal wind u_E via the thermal wind relationship. Here, Δ_h gives the fractional pole to equator temperature difference, Δ_v the fractional temperature difference between the surface and the top of the fluid, H the depth of the fluid, and θ_0 the global mean value of θ_E . This solution not only produces westerly winds at the equator, which are not observed in the atmosphere, but also violate Hide's theorem (Hide, 1969). Hide's theorem states that a maximum of angular momentum within the flow domain cannot be sustained due to dissipation within the flow. A maximum of angular momentum can only exist at the boundaries. This tells us that this RE temperature profile is not able to persist on Earth and that we require some poleward transport of energy.

The conservation of angular momentum

$$M = (u + \Omega a \cos \varphi)a \cos \varphi \quad (2.3)$$

poses another condition for a HC solution with the zonal wind u , a being the radius and Ω the rotational frequency of the Earth. The angular momentum conserving wind can be derived to

$$u_M = \Omega a \frac{\sin^2 \varphi}{\cos \varphi}. \quad (2.4)$$

This allows us to infer, via the thermal wind relationship, a temperature profile θ_M (see Figure 2.1a), which is now slaved to angular momentum conservation. Assuming that the energy is conserved in the upper branch and that the solution is continuous at the edge of the HC, one obtains a finite width of the HC

$$\phi_H \sim \left(\frac{gH\Delta_h}{\Omega^2 a^2} \right)^{1/2} \quad (2.5)$$

and a scale for its strength

$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2}(H\Delta_h)^{5/2}}{\Delta_v a^2 \Omega^3} \quad (2.6)$$

with g being the gravitational acceleration of Earth and τ the time scale at which θ relaxes to θ_E .

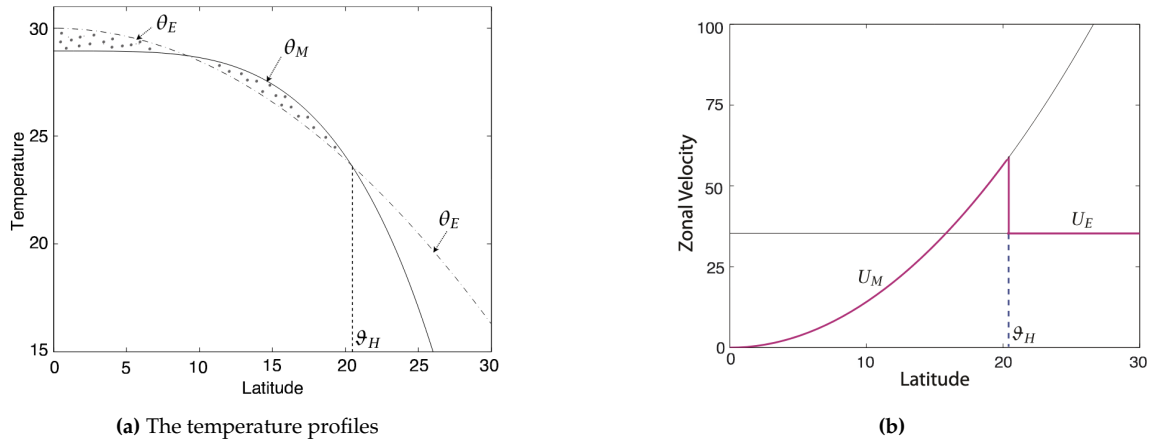


Figure 2.1: (Vallis, 2017)

The contribution of this theory is to explain why there is a HC with finite width in the first place. Subsequent studies have advanced this theory also to temperature profiles which have their maximum off the equator to account for seasonal variations (Lindzen & Hou, 1988). HC weaker in equinox conditions than observed. Suggestion that the HC in the annual mean is not close to equinox conditions but rather that the HC flips between solstitial states. More recent studies have explained the discrepancy using eddy permitting models (Walker & Schneider, 2005). The strength of an annually averaged Hadley circulation is comparable to the Hadley circulation driven by an annually averaged heating. This highlights the importance of eddies to drive the HC strength, which will be covered in the next section.

2.3.2. Eddy driven mean circulation

Early studies about eddies and the mean circulation Eddies can drive a circulation (Ferrel cell) or as in the case of the HC significantly enhance it **Held, baroclinicity, and the HC edge.** **Walker.Schneider,Singh, HC strength and eddies** equinox+summer HC in eddy driven regime winter HC in angular momentum conserving regime

Applying a Reynolds decomposition on the zonal and meridional wind of the zonal momentum equation and ignoring viscous forces as well as vertical advection we obtain

$$f\bar{v}(1 - Ro) = \frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p}. \quad (2.7)$$

The Rossby number is

$$Ro = \frac{\bar{\xi}}{f} \quad (2.8)$$

where $\xi = \hat{k} \cdot \nabla \times \vec{u}$ is the relative vorticity. This equation has two limit: For $Ro \rightarrow 1$, the eddy term on the RHS becomes negligible and our Hadley circulation is close to angular

momentum conservation

$$\frac{\partial \bar{M}}{\partial \varphi} = 0 \quad (2.9)$$

which is described by the axisymmetric solution in Section 2.3.1. For $Ro \rightarrow 0$, the meridional wind is balanced by the eddy momentum flux convergence. This regime can be understood as eddy-driven. During the seasonal cycle the Rossby number varies, which determines how close to the angular momentum conserving limit the HC is. During the equinoxes the Rossby numbers are small which indicates a strong eddy influence. During the solstices the Rossby number is closer to 1 in the winter HC but still small in the summer HC. Assuming that the HC is entirely eddy driven ($Ro = 0$), the HC strength can be predicted from the eddy momentum flux by inserting Equation (2.7) into the Equation (2.1). Similar to Walker and Schneider (2006), this gives

$$\psi_{\max} = \frac{2\pi a \cos \varphi}{fg} \int_{p_{\max}}^0 \left(\frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p} \right) dp' \equiv \mathcal{T} \quad (2.10)$$

with $p_{\max}$ being the pressure level at which the MSF becomes maximum and the Coriolis parameter f .

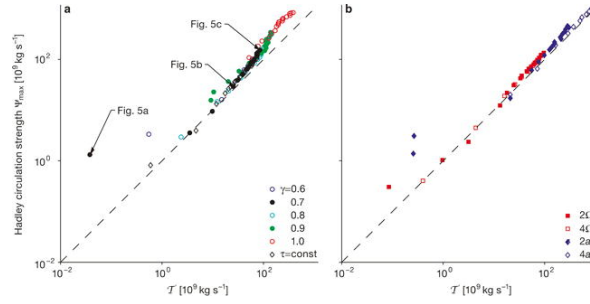


Figure 2.2: (Walker & Schneider, 2006)

2.4. The energy budget of the atmosphere

The energy budget analysis provides diagnostic perspective on the HC. It cannot explain why the Earth is not in radiative (convective) equilibrium. Using TOA radiative fluxes, one can infer that the short-wave radiative flux at the equator is much larger than the emitted outgoing long wave radiation. The opposite is true for the poles. This implies an energy transport by atmosphere and ocean. The atmospheric component can be further split up into a mean and an eddy component.

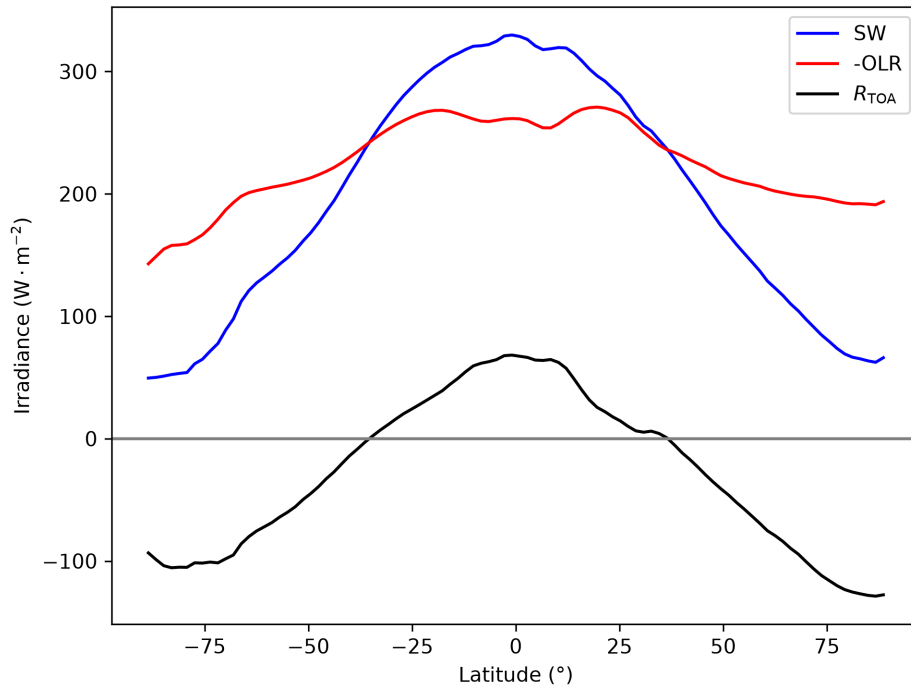


Figure 2.3

$$R_{\text{TOA}} = \frac{\partial F}{\partial y}$$

$$F_{ao} = \int_{-\pi/2}^{\phi} \int_0^{2\pi} R_{\text{TOA}} a^2 \cos \phi d\lambda d\phi$$

$$F_{\text{HC}} = \psi_{\text{max}} H$$

$$H \sim \frac{1}{\psi_{\text{max}}} \int_0^{p_s} [\bar{v}][\bar{h}] dp$$

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{HC}}}{F_{\text{HC}}} - \frac{\delta H}{H}$$

$$F_{\text{HC}} = F_a - F_e$$

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_a}{F_{\text{HC}}} - \frac{\delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H}$$

$$\Delta R_{\text{TOA}} = \sum_i \lambda_i \Delta T_s + R_f$$

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{forc}}}{F_{\text{HC}}} + \frac{\delta F_{\text{fdb}}}{F_{\text{HC}}} - \frac{\delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H}$$

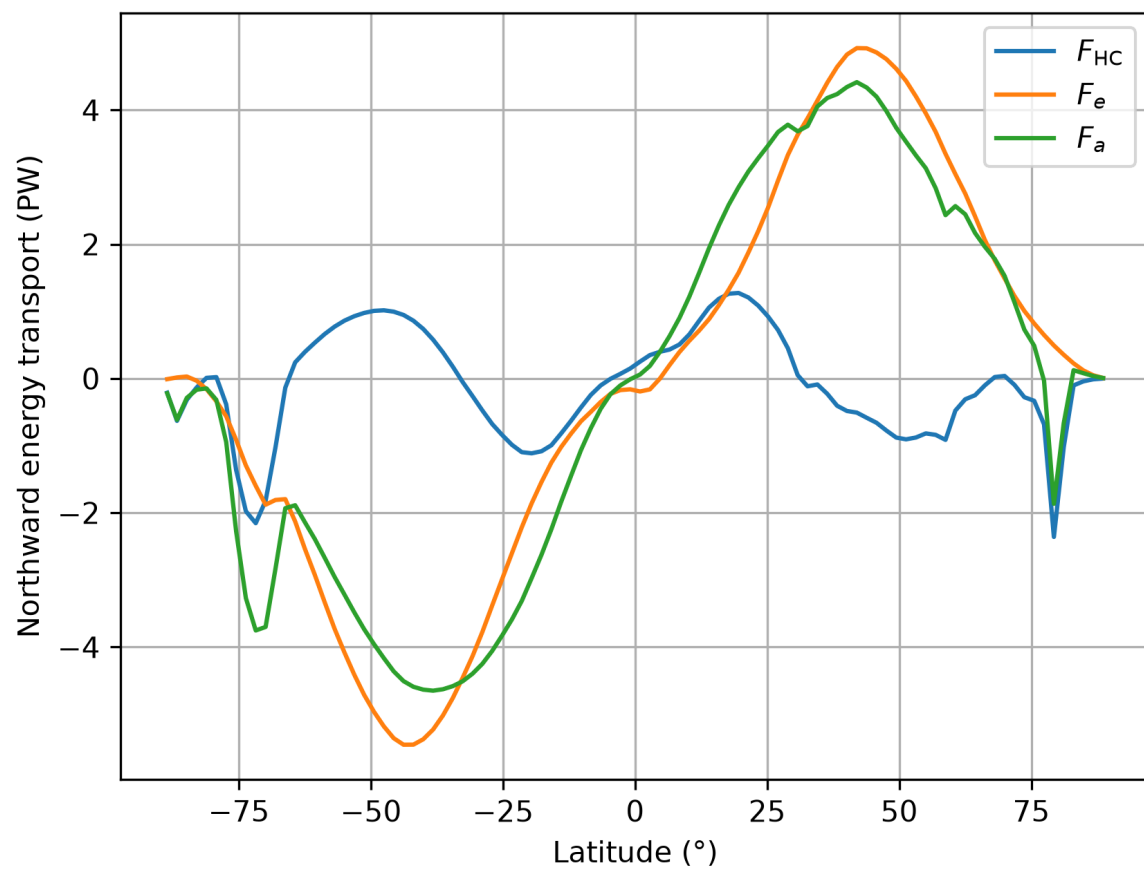


Figure 2.4

2.5. Model setup

2.5.1. ECHAM5

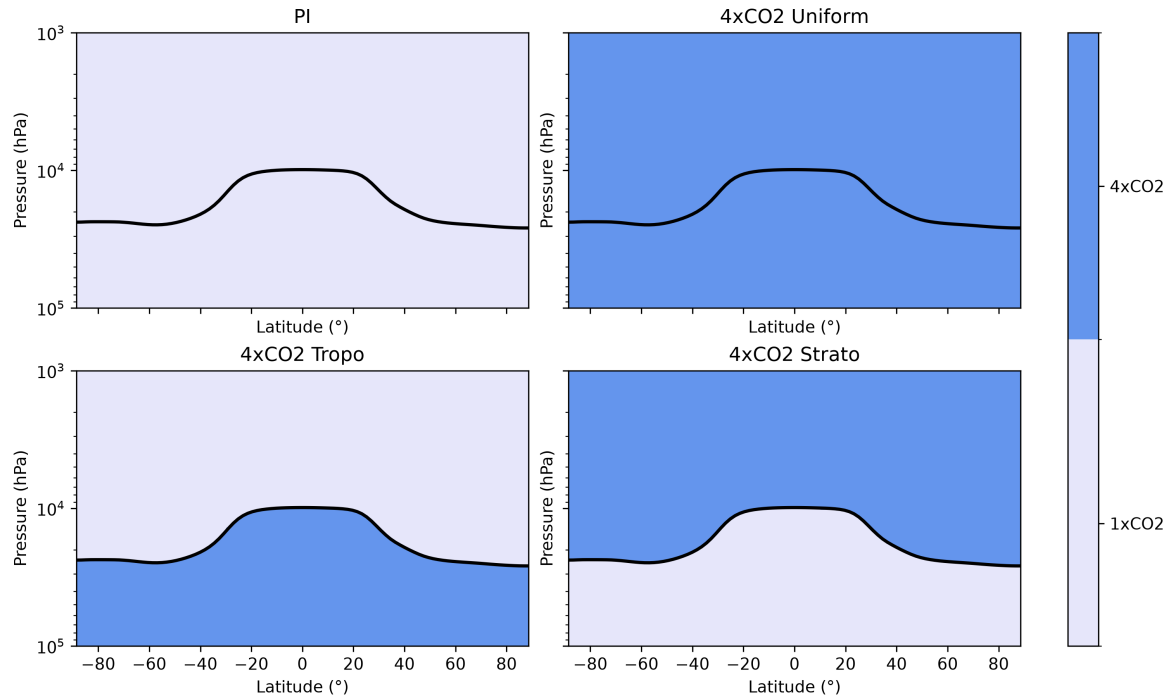


Figure 2.5

2.5.2. konrad

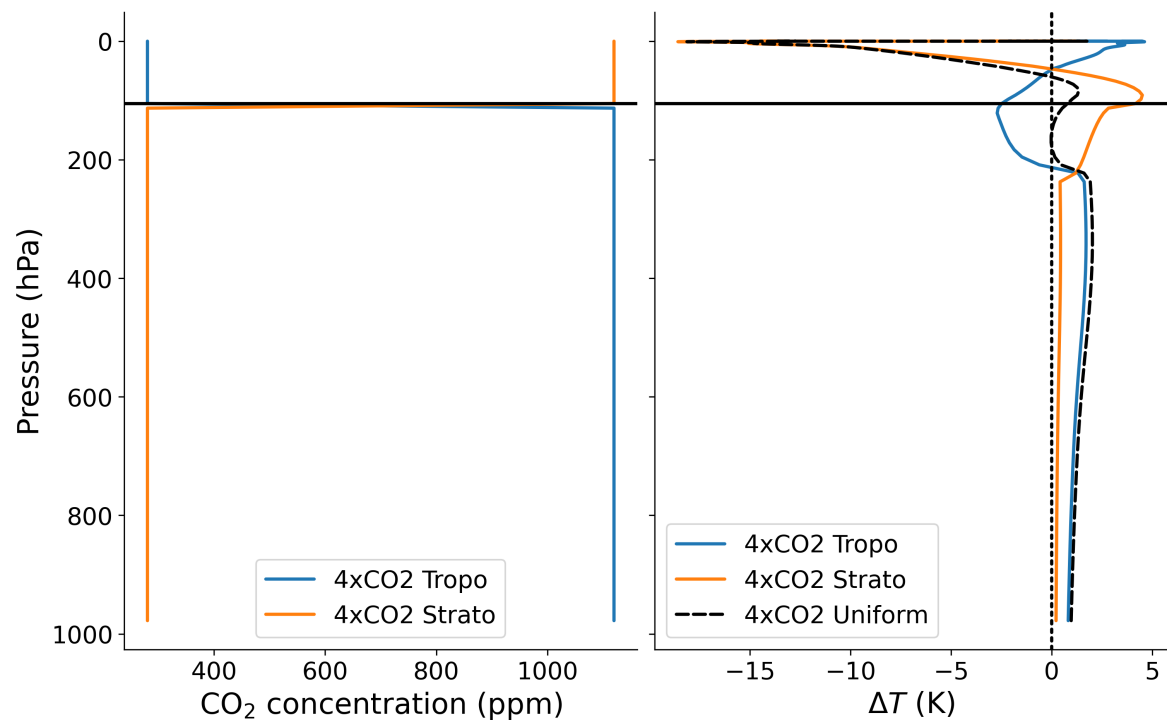


Figure 2.6

3

Methods

- 3.1. ECHAM5
- 3.2. konrad
- 3.3. Decomposition in forcing factors

4

Results

4.1. The atmospheric response

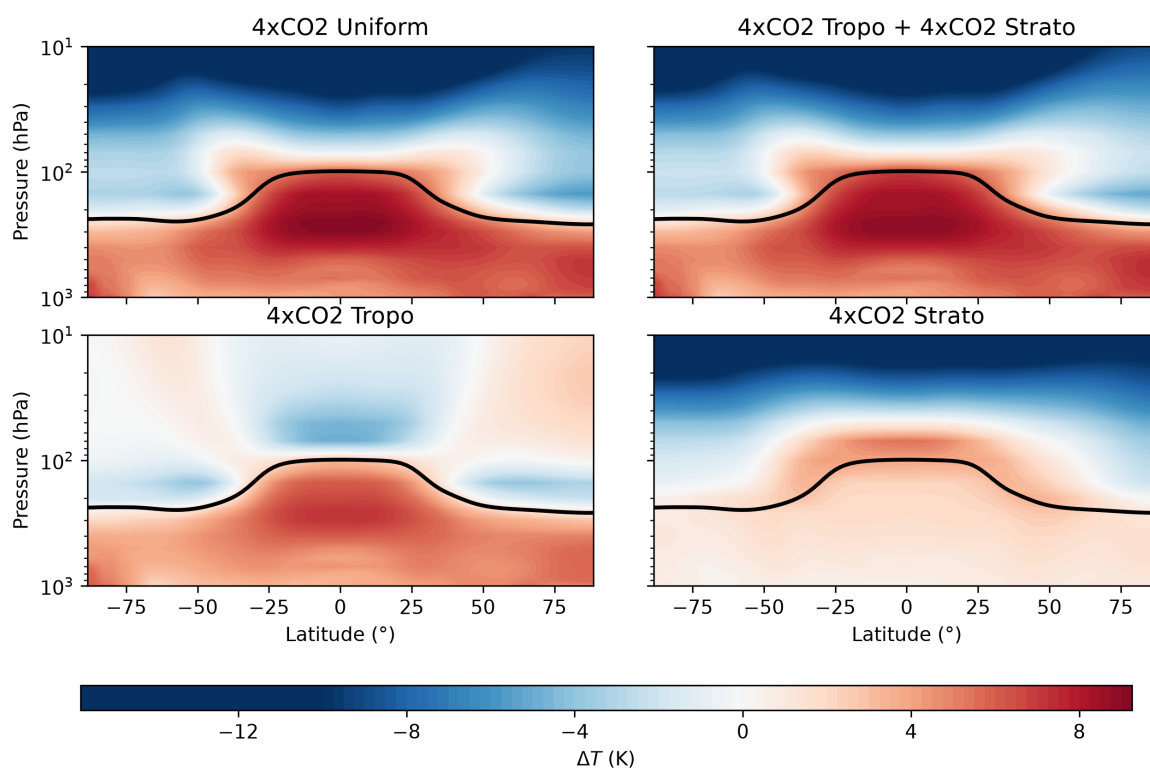


Figure 4.1

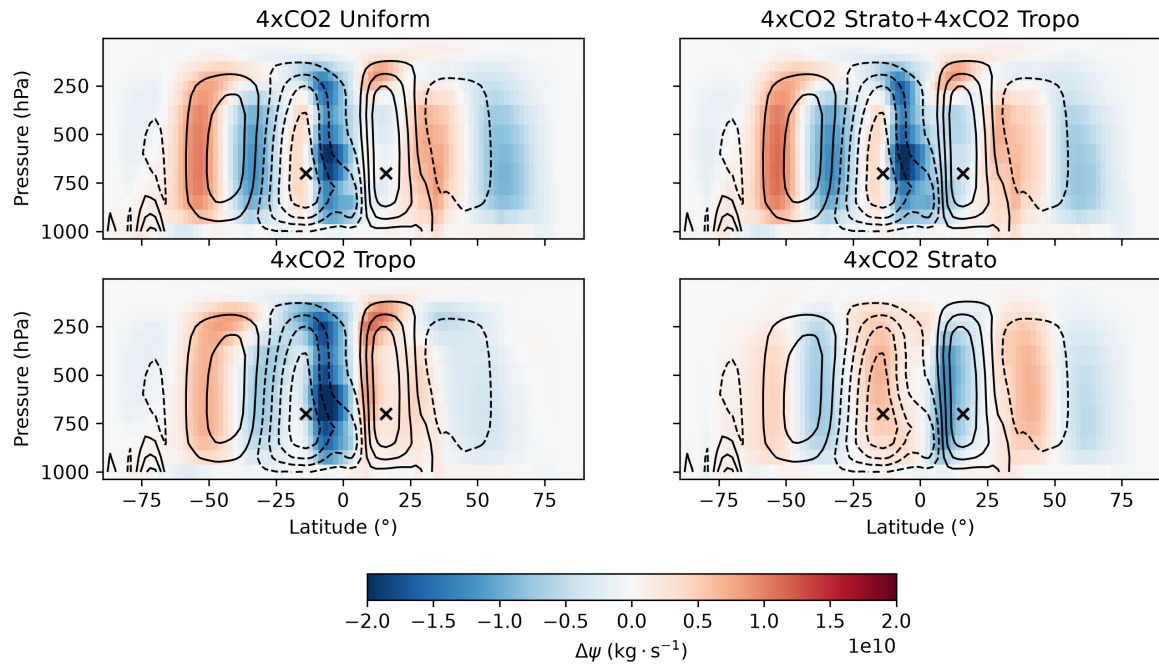


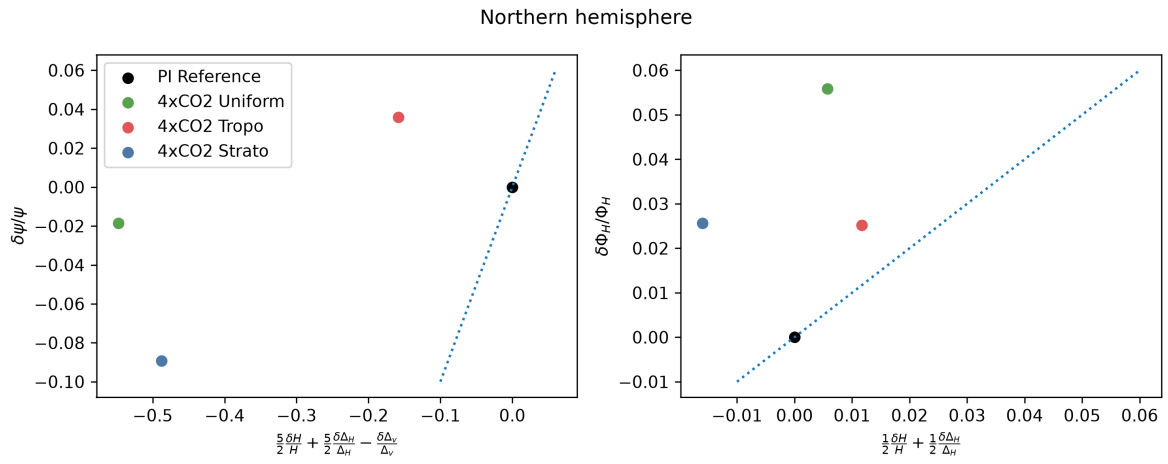
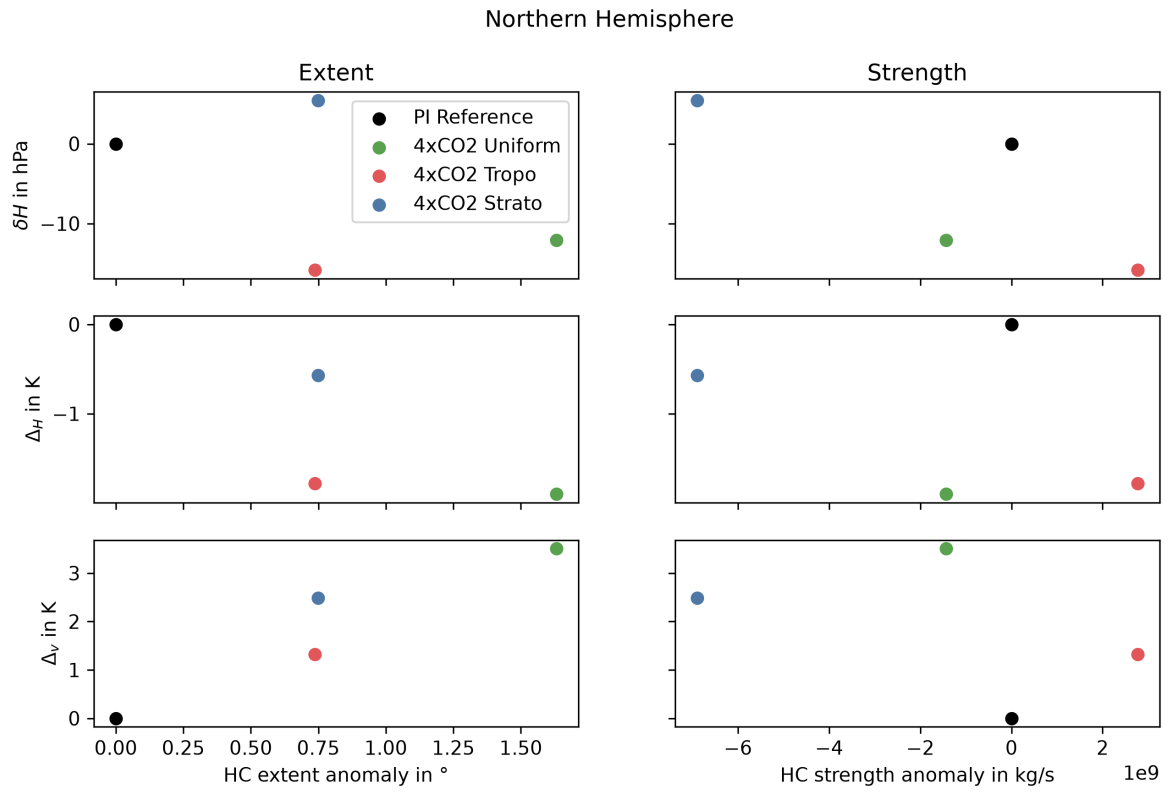
Figure 4.2

4.2. Predictions from axisymmetric scalings

(Seo, Frierson, & Son, 2014)

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{5}{2} \frac{\delta H}{H} + \frac{5}{2} \frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v} \quad (4.1)$$

$$\frac{\delta\phi_H}{\phi_H} = \frac{1}{2} \frac{\delta H}{H} + \frac{1}{2} \frac{\delta\Delta_h}{\Delta_h} \quad (4.2)$$



4.3. Eddy driven Hadley cell predictions

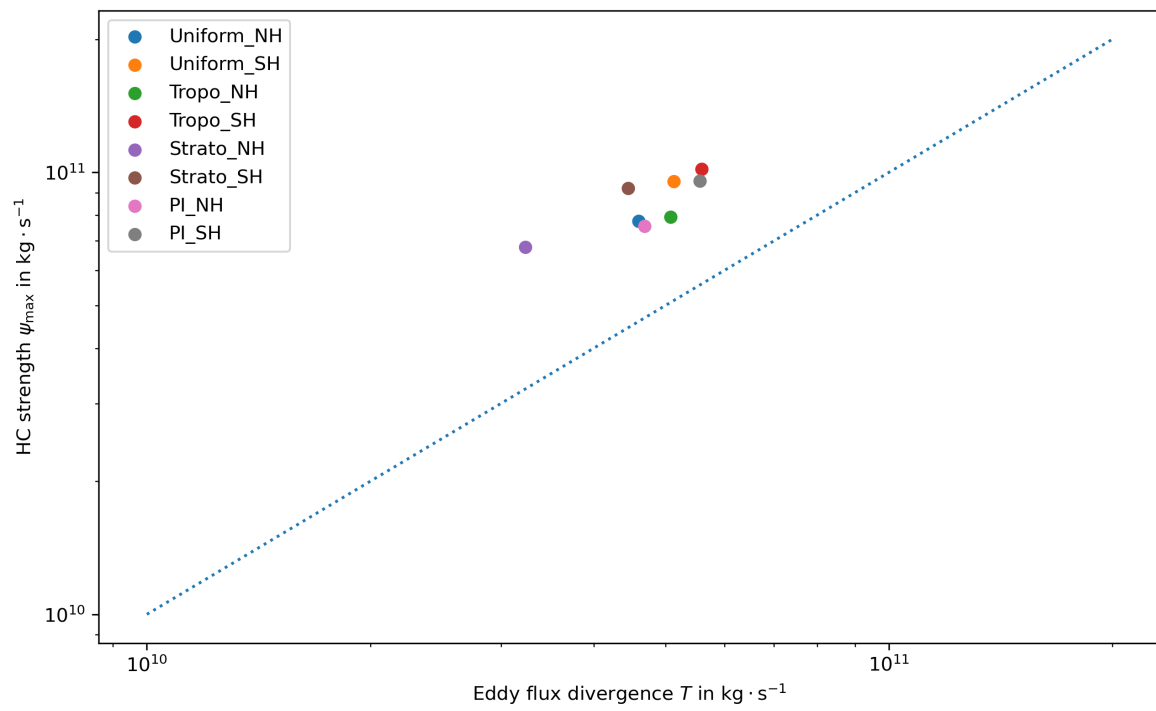


Figure 4.5

4.4. The energetic framework

4.4.1. GMS and energy transport of the Hadley cell

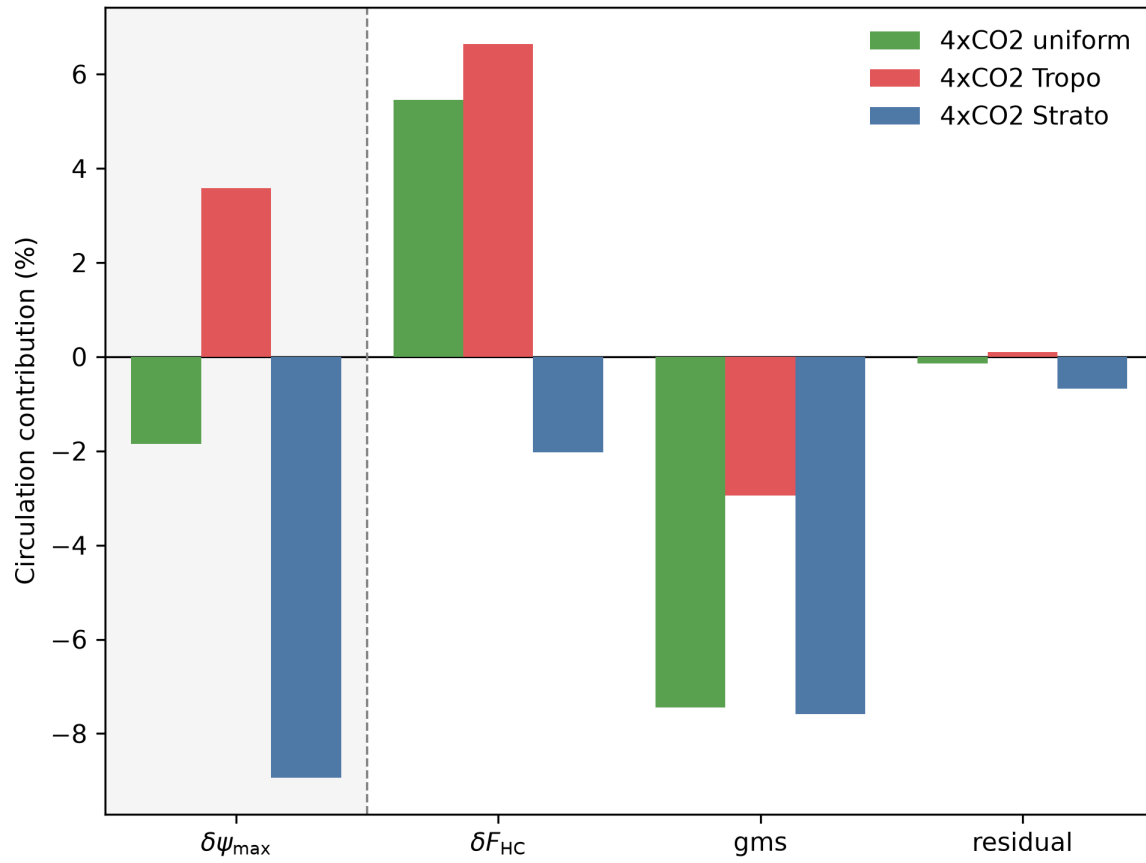


Figure 4.6

4.4.2. Eddy energy transport

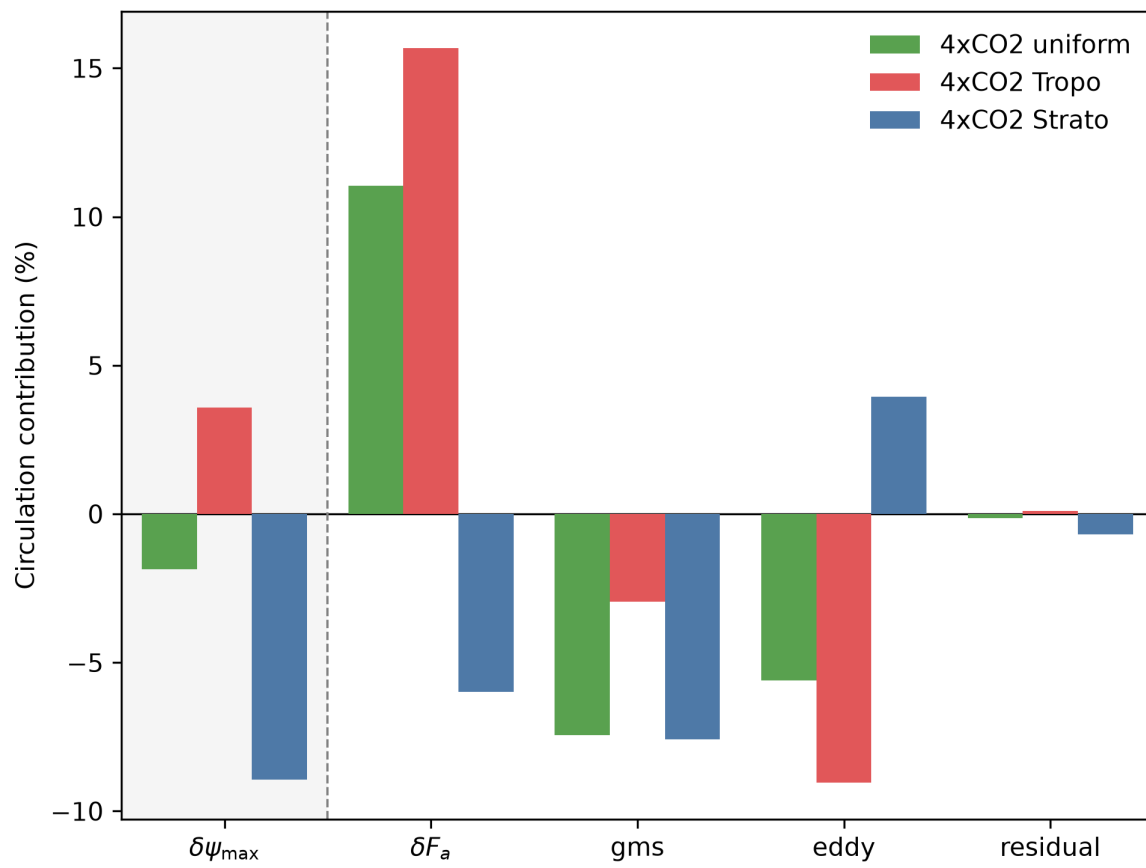


Figure 4.7

4.4.3. Forcing and feedback contributions

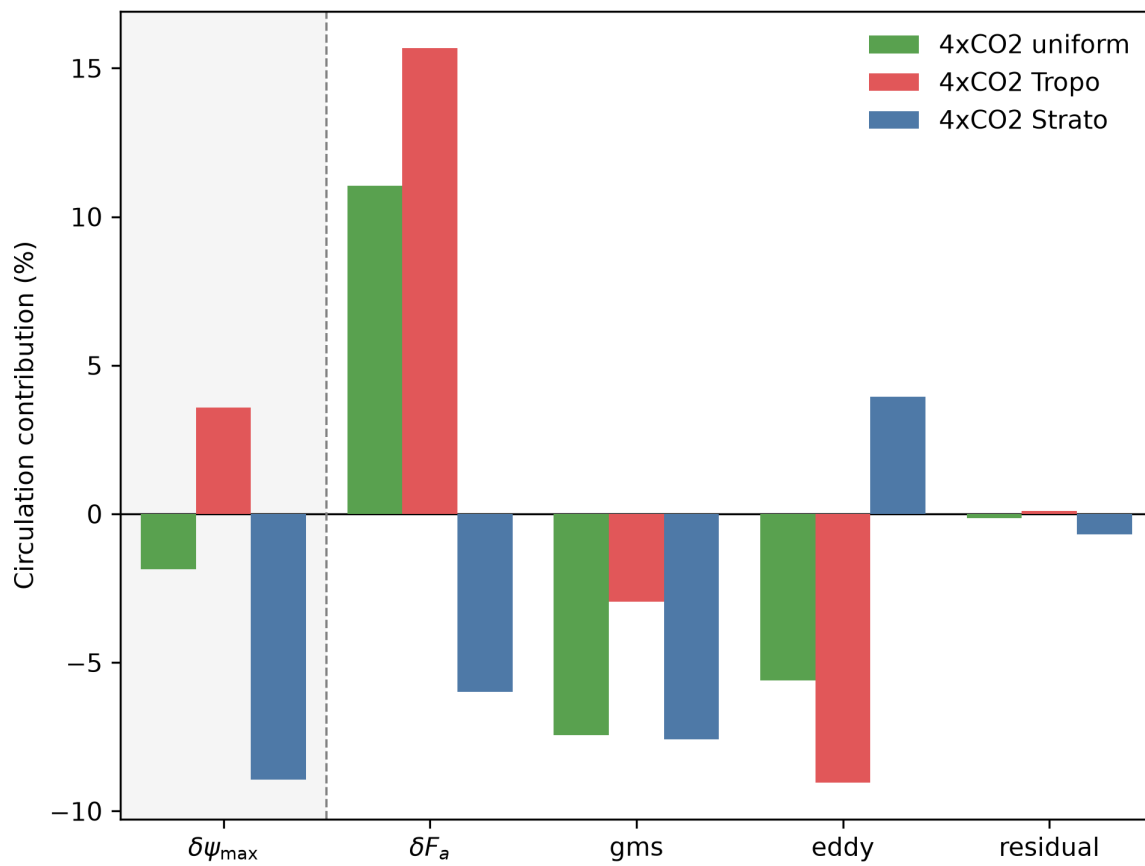


Figure 4.8

4.4.4. The forcing

Explain the forcing structure (Merlis, 2015)

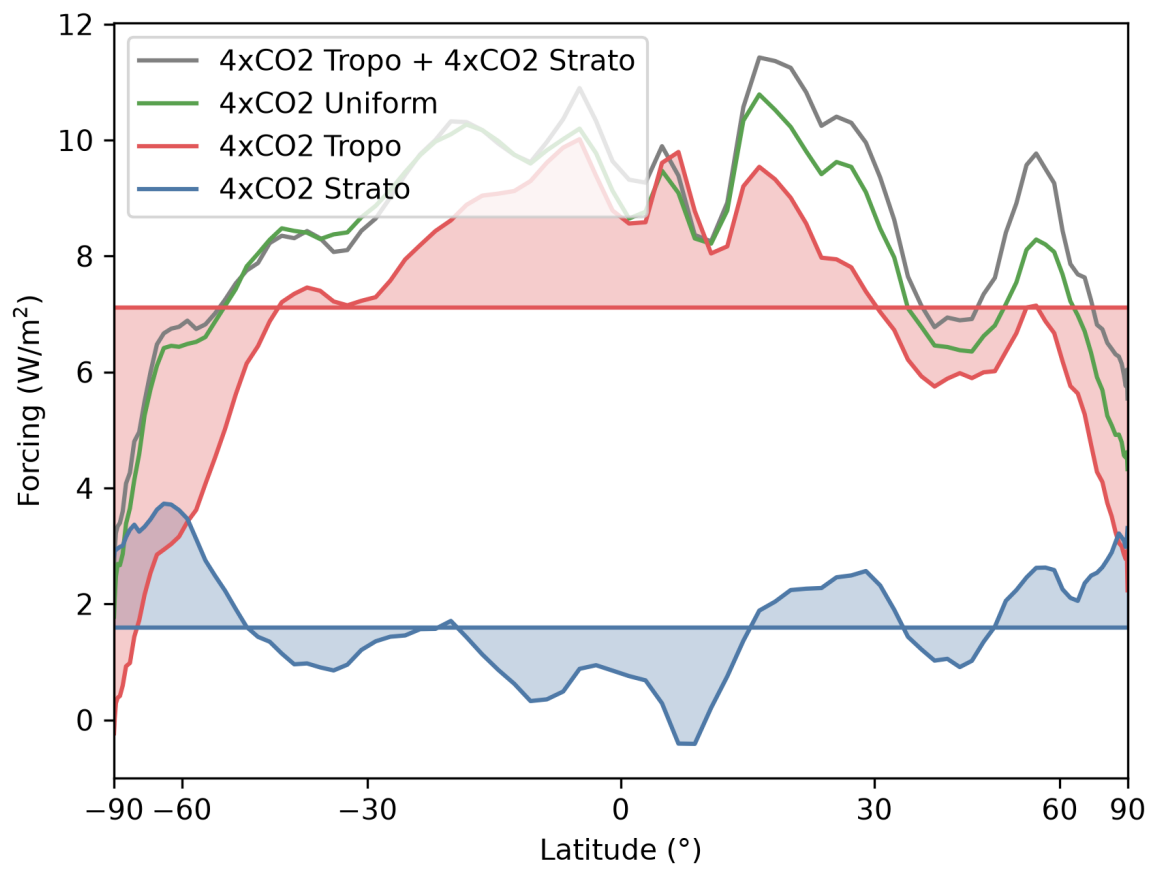


Figure 4.9

5

Discussion

6

Conclusion

Acknowledgement

First, I like to thank my first supervisor , for Also, I want to thank my second supervisor for

Thanks to for all the insights into Thanks to for all the intense but fruitful scientific debate about Thanks to ... , who always had an open door for

Also, I would like to thank the

And last but not least, thanks to , who helped As well as for the help with

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